

REFT Academic Presentation Outline

30-Minute Talk + Discussion Appendix

This document contains a complete strategic presentation outline for the paper: **“REFT: Reasoning with REinforced Fine-Tuning”** (ACL 2024). It includes: Slide-by-slide structure Timing recommendations Visual suggestions Presentation strategy Discussion appendix ideas Academic framing guidance Delivery advice and rhetorical hints The presentation is designed for a 30-minute academic conference-style talk followed by a 15-minute discussion.

Overall Presentation Strategy

Your presentation should feel like: **“Here is a fundamental shift in how we train reasoning models.”** NOT: “Here is another optimization trick.” The central narrative of the presentation should be: Transition from imitation → reasoning optimization Transition from static supervision → exploration-based learning Reasoning as policy optimization rather than behavior cloning Audience emotional flow: Motivation → “This is important.” Problem with SFT → “Current methods are limited.” ReFT insight → “Exploration is missing.” Experiments → “This actually works.” Reward hacking → “This reveals deep RL issues.” Broader impact → “This paper matters beyond math reasoning.”

Presentation Timing Structure

Section	Time
Motivation + Problem	5 min
Background + Prior Work	4 min
ReFT Key Idea	4 min
Method Deep Dive	7 min
Experiments + Results	6 min
Critical Analysis + Weaknesses	3 min
Conclusion + Future	1 min

Slide 1 — Strong Opening

Goal: Immediately capture attention and establish the core research question. **Suggested opening line:**

“If you ask two humans to solve the same math problem, they may use completely different reasoning paths — yet both arrive at the correct answer. But most language models today are trained to imitate only ONE reasoning path.”

Main message: Can language models learn reasoning through exploration rather than imitation? **Visual**

suggestions: One math problem with multiple human solution paths Contrast between human flexibility and static

SFT training Minimalistic, clean title slide **Hints:** Do not begin with “Today I will present...” Sound curious and

intellectually excited Set up the “imitation vs optimization” narrative immediately

Slide 2 — Why Reasoning Matters

Goal: Establish importance. Topics: Reasoning in LLMs Math problem solving Agents and planning Scientific

reasoning Code generation **Visual suggestions:** Icons or examples from coding, planning, mathematics Simple

visual showing reasoning as a multi-step process **Main transition:** Current CoT training is fundamentally

imitation-based.

Slide 3 — Problem with Standard SFT

Goal: Clearly explain why SFT is insufficient. Key ideas: SFT learns one annotated trajectory Reasoning tasks have multiple valid trajectories SFT behaves like behavior cloning Weak generalization can emerge **Visual suggestions:** One question branching into multiple reasoning paths Highlight only one path being supervised in SFT **Important hint:** This is one of the most important conceptual slides in the talk.

Slide 4 — Prior Approaches

Keep concise. Mention: Chain-of-Thought prompting Program-of-Thought / PAL Self-consistency RLHF **Main transition statement:**
“None of these methods directly optimize reasoning behavior itself.”

Slide 5 — Core Idea of ReFT

This is the most important slide in the presentation. Title suggestion: “Optimize Reasoning Instead of Imitating It” Comparison table: **SFT** ReFT Imitate one CoT Explore many CoTs Static supervision Online exploration Token prediction Task reward optimization **Main narrative:** Reasoning becomes a reinforcement learning problem.

Slide 6 — High-Level Pipeline

Pipeline: Pretrained Model → SFT Warm-up → PPO Exploration → Better Reasoning Policy **Visual suggestions:** Simple clean pipeline diagram Avoid clutter Minimal text **Main explanation:** Warm-up provides basic reasoning competence, while PPO enables exploration and policy improvement.

Slide 7 — Why RL Fits Reasoning

Main idea: Reasoning is fundamentally a search process. Key discussion points: Reasoning is not memorization Multiple valid trajectories exist RL naturally supports exploration Policy optimization aligns better with reasoning **Hint:** This is where your presentation becomes intellectually deeper.

Slide 8 — RL Formulation

Keep this technical but intuitive. Explain: **RL Concept** ReFT **Equivalent** State Question + generated tokens Action Next token Policy Language model Reward Answer correctness **Visual suggestion:** Show token-by-token generation process.

Slide 9 — PPO Intuition

Do NOT deeply derive PPO mathematically. Explain conceptually: PPO explores trajectories Rewards successful reasoning KL regularization stabilizes learning PPO prevents unstable policy updates **Hints:** Use intuition over formulas One small PPO diagram is enough Avoid equation-heavy slides

Slide 10 — N-CoT vs P-CoT

Show side-by-side examples. Discuss: Natural language reasoning Program-based reasoning Explain why P-CoT performs better: Executability Symbolic consistency Lower ambiguity Harder to fake

Slide 11 — Experimental Setup

Keep concise. Mention: Datasets: GSM8K, SVAMP, MathQA Models: Galactica, CodeLLaMA N-CoT and P-CoT settings **Hint:** Do not spend too much time on setup details.

Slide 12 — Main Results

Highlight only the most important gains. Focus: ReFT significantly outperforms SFT Large gains on GSM8K P-CoT consistently stronger **Very important interpretation:**
“These improvements occur WITHOUT additional training data.”

Slide 13 — Why ReFT Works

Key ideas: Exploration matters Learning from failures matters Trajectory distributions are richer RL optimizes reasoning behavior directly **Main insight:** SFT learns one trajectory; ReFT learns a distribution of trajectories.

Slide 14 — Reward Hacking

Use the rhombus example from the paper. Explain: Reasoning is wrong Final multiple-choice answer is correct Reward becomes positive despite bad reasoning **Main interpretation:**
This is a classic reinforcement learning problem: reward misspecification. **Hint:** This slide will strongly impress professors if explained well.

Slide 15 — Why Reward Hacking Matters

Connect the issue to broader AI topics: AI alignment Specification gaming RL safety Reward misspecification **Goal:** Demonstrate that the paper touches deeper RL problems beyond math reasoning.

Slide 16 — Broader Impact on LLM Research

Connect ReFT to: RLVR DeepSeek-R1 style training Verifier-based RL Process Reward Models Reasoning agents
Main message: This paper helped push the field from imitation-based reasoning toward RL-based reasoning optimization.

Slide 17 — Strengths and Weaknesses

Strengths: Elegant idea No reward model required Strong empirical gains Online exploration Same data, better optimization **Weaknesses:** Sparse rewards Reward hacking PPO instability Expensive training Outcome-only reward function

Slide 18 — Personal Critique

This slide is extremely important academically. Suggested critique: The paper reframes reasoning as policy optimization rather than imitation. Outcome-only rewards are too weak for robust reasoning supervision. The paper foreshadows later RL reasoning systems. Historically important transition paper in RL-for-reasoning research. **Hint:** Sound thoughtful, analytical, and engaged with the field.

Slide 19 — Future Work

Mention: Process Reward Models (PRMs) Verifier-guided RL Tool-assisted reasoning Step-level supervision Reasoning agents Potential references: DeepSeek-R1 Let's Verify Step by Step AgentPRM

Slide 20 — Final Takeaway

Strong closing message: "ReFT demonstrates that reasoning ability may emerge more effectively from exploration and reward optimization than from pure imitation." **Final emotional takeaway:** This paper represents a broader transition in AI: from supervised imitation toward reinforcement-driven reasoning optimization.

Backup / Appendix Slides

Prepare approximately 8–12 appendix slides. Suggested appendix topics: PPO details and clipping intuition Why PPO instead of DPO KL regularization explanation Reward hacking deeper analysis Why P-CoT works better Relation to

RLHF Why warm-up is necessary Training stability Modern follow-up works **Hint:** Backup slides strongly improve discussion performance and confidence.

Delivery Advice

Important presentation guidance: Do not sound like you are reading slides. Continuously return to the central narrative: "Reasoning should be optimized, not merely imitated." Use enthusiasm and intellectual curiosity. Focus on intuition and implications rather than excessive formulas. Do not overload slides with text. Use clean visuals and consistent color themes. **Most important slides to perfect:** Problem with SFT Core Idea of ReFT Why ReFT Works Reward Hacking Personal Critique